

SMRITI SHARMA

Python Backend Engineer | AI/ML Engineer | Data-Intensive Systems

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PROFESSIONAL SUMMARY

Python-focused Software Engineer with 4+ years of experience in software engineering, backend automation, REST APIs, CI/CD pipelines, and data validation systems. Recently completed a Master's degree in Computer Science with Machine Learning specialization. Experienced in building reliable Python-based systems, integrating APIs and databases, writing automated tests, and supporting scalable data-driven applications. Strong background in software quality, debugging, agile collaboration, and translating complex technical requirements into maintainable solutions.

WORK EXPERIENCE

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| Student Research Assistant (HiWi),
<i>German Research Center for Artificial Intelligence (DFKI)</i> | 10/2024 – 03/2026
Kaiserslautern,
Germany |
| <ul style="list-style-type: none">• Developed an LLM-powered intelligent coding assistant using Javascript that generated contextual hints and explanations using GPT-based reasoning• Designed backend workflows and retrieval mechanisms for processing user queries and generating context-aware responses• Evaluated prompt strategies and response quality to improve user experience and system reliability• Collaborated with researchers to translate AI concepts into deployable software prototypes | |
| Student Research Assistant (HiWi),
<i>German Research Center for Artificial Intelligence (DFKI)</i> | 06/2023 – 03/2024
Kaiserslautern,
Germany |
| <ul style="list-style-type: none">• Built data processing pipelines for ML applications, including dataset preparation and validation• Created dataset from scratch for wearable AI validation• Supported development and evaluation of AI models | |
| Software Development Engineer in Test, Akamai | 07/2021 – 09/2022
Bangalore |
| <ul style="list-style-type: none">• Developed Python-based automation frameworks for API and backend testing in distributed systems, improving test coverage by 30–40%.• Designed CI/CD pipelines using Jenkins and Azure DevOps for automated build and deployment workflows• Implemented automated testing strategies (unit, integration, system testing) for distributed systems• Improved test execution efficiency by ~35% through optimization of pipelines | |
| Senior Software Engineer, Mindtree Ltd. | 10/2018 – 06/2021 |
| <ul style="list-style-type: none">• Built Python-based data validation pipelines comparing backend database records, REST API responses, and frontend data representations, ensuring >95% data consistency• Developed UI and API automation suites using Ruby and Python for enterprise applications, reducing manual regression effort by >50%.• Integrated test frameworks into Jenkins CI/CD pipelines for continuous validation and release readiness | |

- Worked with SQL databases and data workflows to identify transformation issues, backend inconsistencies, and application-level defects
- Created and maintained reusable test environments, reducing environment-related issues by ~25%

EDUCATION

Master of Science – Computer Science, RPTU KAISERSLAUTERN	10/2022 – 03/2026
Grade: 2.2	Germany
Bachelor of Technology - Computer Science & Engineering, ITER	2014 – 2018
Grade: 9.10/1.6	INDIA

TECHNICAL SUMMARY

Programming: Python, JavaScript, Shell	Languages:
AI / Machine Learning: PyTorch, TensorFlow, Transformers, NLP, Computer Vision, Deep Learning	English: Advanced German: Beginner (A2)
Data: PostgreSQL ,pgvector, Pandas, NumPy, Data Pipelines, ETL Pipelines, ChromaDB	Tools & Platforms: GitHub, Bitbucket, Jira, Kanban Boards, Bash, Postman, PyCharm, Eclipse, VS Code, Jupyter Notebook, pgAdmin, Tableau, Excel
Backend Engineering: REST APIs, Flask/FastAPI basics, SQL, API testing, Backend workflows	DevOps: Git, Jenkins, Azure DevOps, Docker, CI/CD, Linux
Testing: Pytest, Unit Testing, integration testing, System Testing, Regression Testing, UI/API Testing, Data Validation	Software Practices: Clean code, Debugging, Agile development, System reliability, Documentation

PROJECTS

Master Thesis,	03/2025 – 09/2025
<i>From Prediction to Retention: Transforming Web Agents with Memory Systems.</i>	
• Designed and implemented a memory-augmented LLM web agent by extending WebCanvas with workflow memory, webpage caching, and URL generalization for multi-step web automation tasks • Built a PostgreSQL + pgvector-based memory store to persist task histories, metadata, DOM observations, and reusable workflows using SentenceTransformer embeddings and semantic retrieval • Developed Python evaluation pipelines to benchmark agent performance on Mind2Web-Live, analyzing task success, failures, productivity, token usage, and cost-efficiency • Improved unseen-task performance by ~59% in task success rate and ~50% in productivity, while reducing task failures by ~17% through workflow reuse and reduced redundant reasoning	
	Grade: - 1.0
Master Project, - Master's Project – SparseBEV for Multi-Sensor Fusion	10/2023 – 03/2024
• Implemented feature extraction pipeline using PyTorch for 3D object detection on the nuScenes dataset • Worked with ResNet50 and Feature Pyramid Networks (FPN) for multi-scale feature representation • Processed multimodal data (camera + LiDAR) for multi-sensor fusion tasks	